GYANMANJARI INNOVATIVE UNIVERSITY

Gyanmanjari Institute of Technology – Faculty of Engineering &Technology

Unsupervised Learning and Dimensionality Reduction Case Study

K-Means, Kernel K-Means, PCA and Kernel PCA on the Wine chemistry data

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Activity	ALA 2	Marks	10

Deliverables: the executed Jupyter notebook `ML_ALA2_Unsupervised_Learning_Notebook.ipynb` (code, outputs and explanations) and this brief explanation of algorithm selection, results and conclusions.

1. Aim and Dataset

To apply **unsupervised learning** techniques (K-Means, Kernel K-Means) and **dimensionality reduction** (PCA, Kernel PCA) to a given dataset, analyse the clustering behaviour, visualise the results and interpret the reduced feature space, i.e. to understand the structure of the data *without labelled outputs*.

Dataset: *Wine* (UCI, included in scikit-learn): 178 wines described by 13 chemical measurements (alcohol, flavanoids, colour intensity, proline, …) from three grape cultivars. The cultivar label is **not used for learning**; it is used only afterwards to check the discovered clusters (external validation with the Adjusted Rand Index, ARI, and Normalised Mutual Information, NMI).

2. Algorithm Selection

Technique	Reason for selection	Key points
K-Means	Fast, simple partitioning of compact clusters	Needs k and standardised features; minimises within-cluster sum of squares; sensitive to initialisation (use many starts)
Kernel K-Means	Finds non-linearly separable clusters	Works with an RBF kernel matrix K instead of coordinates; implemented from scratch; needs the kernel width gamma
PCA	Remove redundancy between correlated measurements and visualise in 2-D	Directions of maximum variance; loadings show which original features form each component
Kernel PCA	Non-linear version of PCA	Projection in the kernel feature space; gamma must be tuned

3. Results

Preprocessing and choice of k

The features have very different scales (proline reaches about 1700; hue is around 1), so they were standardised. This matters: K-Means on the raw data reached an ARI of only **0.371** because proline dominates the distance, but **0.897** after standardisation.

k	Inertia (elbow)	Silhouette	Davies–Bouldin(loweris better)	Calinski–Harabasz
2	1658.8	0.259	1.526	69.5
3	1277.9	0.285	1.389	70.9
4	1175.4	0.260	1.797	56.2
5	1109.5	0.202	1.808	47.0
6	1046.0	0.237	1.554	41.7
7	981.6	0.204	1.661	38.7
8	935.2	0.157	1.741	35.8

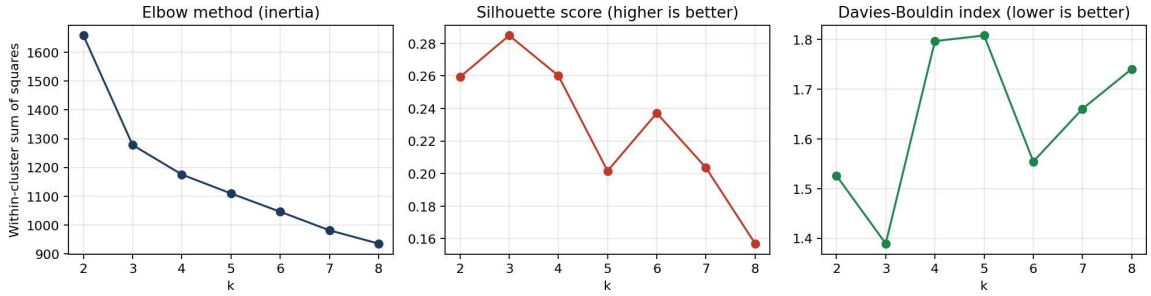


Figure1:Elbow,silhouetteandDavies–Bouldincurves:k=3givesthebestsilhouetteandthelowestDavies–Bouldinvalue, in line with the three cultivars.

With k = 3 the K-Means clusters have sizes [65, 51, 62]. Twenty single random initialisations agree with the best solution on average (mean ARI 0.97, minimum 0.882); a few runs end in a slightly different local optimum, so several starts are used.

PCA: reduced feature space

The first two principal components explain 55% of the variance; 5 components are needed for 80% and 8 for 90%, so the 13 chemical measurements are strongly redundant.

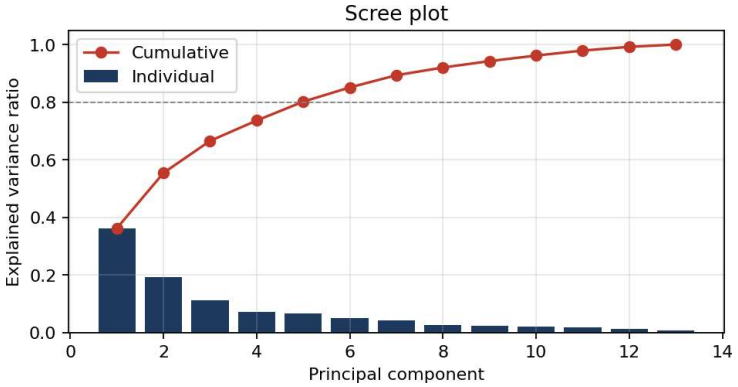


Figure 2: Scree plot (individual and cumulative explained variance).

Component	Strongest loadings (feature: loading)	Interpretation
PC1 (36%)	flavanoids: 0.42, total_phenols: 0.4, od280/od315_of_diluted_wines: 0.38, proanthocyanins: 0.31	**Phenolic content axis**: wines rich in flavanoids,phenolsandproanthocyanins score high
PC2 (19%)	color_intensity: 0.53, alcohol: 0.48, proline: 0.36, ash: 0.32	**Intensity/strengthaxis**:colour intensity, alcohol and proline

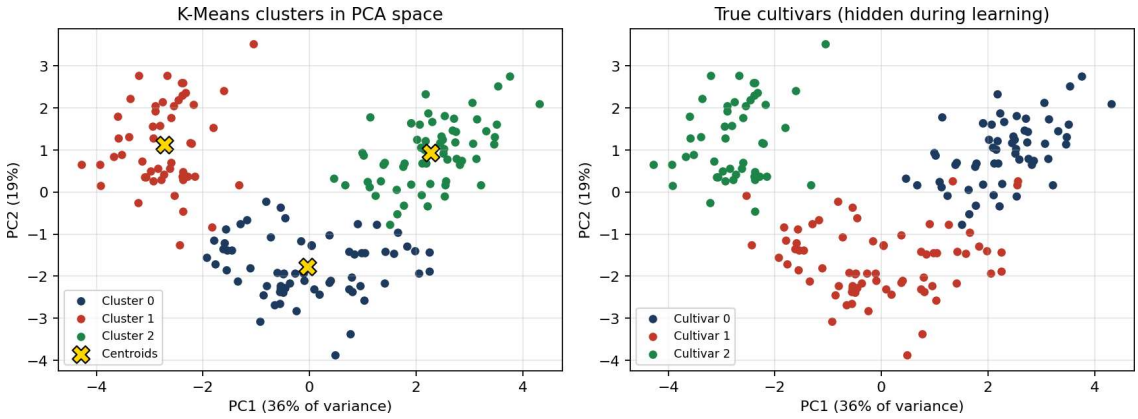


Figure3:K-Meansclusters(left)andthetruecultivars(right)inthePC1–PC2plane;theclustersalmostcoincidewiththe cultivars.

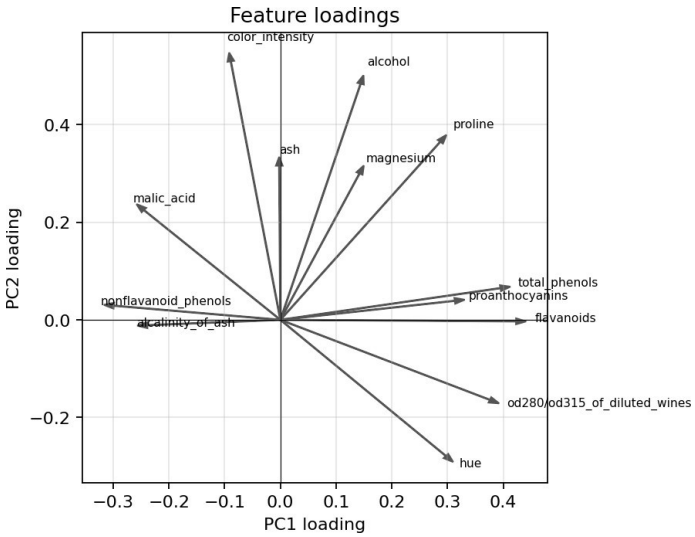


Figure 4: Loadings of the original features on PC1 and PC2.

Kernel methods

RBF gamma	KernelK-Means:cluster sizes	KernelK-Means ARI	Kernel PCA + K-Means: silhouette	KernelPCA+ K-Means ARI
0.002	[62, 65, 51]	0.897	-	-
0.005	[62, 65, 51]	0.897	0.572	0.912
0.01	[51, 61, 66]	0.915	0.582	0.912
0.02	[61, 66, 51]	0.915	0.596	0.913
0.05	[69, 58, 51]	0.933	0.617	0.931
0.1	[58, 50, 70]	0.949	-	-

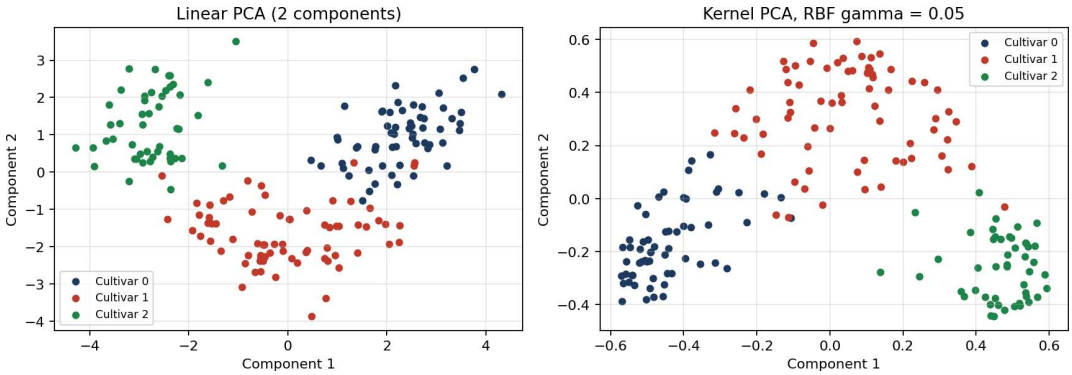


Figure 5: Linear PCA and Kernel PCA projections (coloured by true cultivar).

Comparison of methods

Method	Silhouette(own space)	ARI vs cultivar	NMI vs cultivar
K-Means (13 scaled features)	0.285	0.897	0.876
K-Means on raw (unscaled) data	0.571	0.371	0.429
PCA (2 comp.) + K-Means	0.561	0.895	0.882
Kernel PCA (gamma=0.05) + K-Means	0.617	0.931	0.909
Kernel K-Means (gamma=0.1)	0.282	0.949	0.928

Silhouettevaluesarecomputedindifferentspacesandarenotcomparablewitheachother;ARI/NMIcompareeachresult with the hidden cultivar labels.

4. Interpretation and Conclusions

- **Structure found without labels:** K-Means with k = 3 on standardised data recovered the three cultivars with ARI 0.897; PCA + K-Means gave about the same result in only two dimensions.
- **Why scaling and PCA matter:** distance-based methods are dominated by large-valued features if the data are not standardised; PCA shows that two interpretable axes (phenolic content and intensity) capture the main differences between wines.
- **Kernel methods:** Kernel PCA + K-Means (best ARI 0.931) and Kernel K-Means (best ARI 0.949) agree slightly better with the cultivars than plain K-Means. This gain is optimistic: the kernel width gamma was selected by looking at the hidden labels, the best value is at the edge of the tested range, and the improvement is only a few wines. Since the standardised data are already almost linearly separable, the kernels give at most a small advantage here.
- **Algorithm choice in practice:** start with scaled K-Means and PCA (simple, fast, interpretable); use kernel versions only when the groups are clearly non-linear, and choose gamma by an internalcriterion.
- **Limitations:**asmall,cleandatasetwithwellseparatedgroups;ARI/NMIneedlabelsthatarenormally unavailable, so real projects must rely on internal indices (silhouette, Davies–Bouldin) and domain knowledge.

References

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